Task 1: Guided Activity, Course registration

main.swift

Run ▶

```swift
import Foundation
var subjects = ["Swift", "UIKit", "SwiftUI"]
subjects.append("Networking")
subjects.insert("Testing", at: 1)
if let index = subjects.firstIndex(of: "UIKit") {
    subjects.remove(at: index)
}
print("Sorted subjects:", subjects.sorted())
var registered: Set<String> = ["Anu", "Ben", "Chitra"]
registered.insert("Dev")
registered.insert("Anu")
print("Registered students:", registered.count)
var marks: [String: Int] = [
    "Anu": 85,
    "Ben": 92,
    "Chitra": 78
]
marks["Dev"] = 88
marks["Anu"] = 90
let missingMark = marks["Esha"]
print("Esha's mark:", missingMark as Any)
for name in marks.keys.sorted() {
    print("\(name): \(marks[name]!)")
}
if let topper = marks.max(by: { $0.value < $1.value }) {
    print("Highest mark:", topper.value)
    print("Top student:", topper.key)
}
```

Output

```
Sorted subjects: ["Networking", "Swift", "SwiftUI", "Testing"]
Registered students: 4
Esha's mark: nil
Anu: 90
Ben: 92
Chitra: 78
Dev: 88
Highest mark: 92
Top student: Ben
```

Task 2: Practice Task, Word frequency counter

main.swift

Run ▶

```swift
import Foundation
let sentence = "swift is fun and swift is fast"
let words = sentence.split(separator: " ").map(String.init)
print("Total words:", words.count)
let uniqueWords = Set(words)
print("Unique words:", uniqueWords.sorted())
var frequency: [String: Int] = [:]
for word in words {
    frequency[word, default: 0] += 1
}
print("Word frequencies:")
for word in frequency.keys.sorted() {
    print("\(word): \(frequency[word]!)")
}
let rankedWords = frequency.sorted {
    if $0.value != $1.value {
        return $0.value > $1.value
    }
    return $0.key < $1.key
}
print("Ranked by frequency:")
for item in rankedWords {
    print("\(item.key): \(item.value)")
}
```

Output

```
Total words: 7
Unique words: ["and", "fast", "fun", "is", "swift"]
Word frequencies:
and: 1
fast: 1
fun: 1
is: 2
swift: 2
Ranked by frequency:
is: 2
swift: 2
and: 1
fast: 1
fun: 1
```

Task 3(a): Challenge Task, Set algebra

main.swift Run ▶

```swift
import Foundation
let swiftClub: Set<String> = ["Anu", "Ben", "Chitra", "Dev"]
let uiClub: Set<String> = ["Chitra", "Dev", "Esha"]
let bothClubs = swiftClub.intersection(uiClub)
let onlySwift = swiftClub.subtracting(uiClub)
let everyone = swiftClub.union(uiClub)
let exactlyOne = swiftClub.symmetricDifference(uiClub)
print("Both clubs:", bothClubs.sorted())
print("Only Swift club:", onlySwift.sorted())
print("Everyone:", everyone.sorted())
print("Exactly one club:", exactlyOne.sorted())
```

Output

```
Both clubs: ["Chitra", "Dev"]
Only Swift club: ["Anu", "Ben"]
Everyone: ["Anu", "Ben", "Chitra", "Dev", "Esha"]
Exactly one club: ["Anu", "Ben", "Esha"]
```

Task 3(b): Challenge Task, De-duplication

main.swift Run ▶

```swift
let numbers = [3, 1, 3, 2, 1, 4]
var seen = Set<Int>()
var uniqueNumbers: [Int] = []
for number in numbers {
    if !seen.contains(number) {
        seen.insert(number)
        uniqueNumbers.append(number)
    }
}
print("Original array:", numbers)
print("Without duplicates:", uniqueNumbers)
```

Output

```
Original array: [3, 1, 3, 2, 1, 4]
Without duplicates: [3, 1, 2, 4]
```